

A Resistance Sign-Based Method for Voltage Sag Source Detection

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Abstract—A method to determine the origin of voltage sag disturbance is proposed in this paper. Its principle is to estimate the equivalent impedance of the nondisturbance side by utilizing the voltage and current changes caused by the disturbance. The sign of the real part of the estimated impedance can reveal if the disturbance is from upstream or downstream. The proposed method can be easily implemented into power quality meters for troubleshooting applications or into digital revenue meters to document the responsibility for disturbances. Extensive field and laboratory test results have proven the effectiveness and robustness of the proposed method.

Index Terms—Least-squares method, power quality, voltage sag, voltage sag source detection.

I. INTRODUCTION

VOLTAGE sag is defined as “a decrease to between 0.1 and 0.9 p.u. in rms voltage at the power frequency for a duration of 0.5 cycle to 1 min” [1]. They are one of the most concerned power quality disturbances in power systems. The sources for voltage sags are usually system faults, heavy load-switchings, and large motor startings. Sags can propagate long distance from the source location and affect customers in large areas. Over the past many years, research work in voltage sag diagnosis has been focused on developing protocols and systems to measure the disturbance. Little work has been reported on locating the origins of voltage sags. With the deregulation of power industry, utilities have become increasingly interested in quantifying the responsibilities for power quality problems, with voltage sag as the most prominent concern. As a result, research on how to detect sag source direction has become an important topic recently. In spite of various efforts, there is still no reliable and technically sound method available for locating sag sources.

Reference [2] is probably the pioneering work in this subject. It uses disturbance energy and the initial peak of the disturbance power to determine directions of disturbances. The method is based on intuitive understanding of the disturbance phenomena. Theoretical support for the method is not available. As a result, it is difficult to assess the reliability and limitations of the method.

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Reference [3] proposed a method based on the observation that the slopes of the voltage versus current plot can have opposite signs depending on the sag source location. By examining the sign of the slope, one may be able to tell if a disturbance is located at upstream or downstream. Again, this is an empirical method and it is difficult to assess its application range.

Since voltage sags are mainly related to faults, it would be useful to examine the knowledge accumulated in the area of power system protection and fault locations. We found that the concept of “incremental impedance” used in directional relays is the most relevant to the problem of sag source detection [4]. The method is based on the observation that the estimated impedance obtained from incremental voltage and current quantities at different fault locations will lie in different quadrants of the impedance plane. One of the problems of this method is that the impedance is sensitive to the selections of pre- and post-fault cycles [5]. Different cycle selection may give opposite results.

In this paper, we propose a theoretically justifiable method for sag source detection. The method is inspired by the impedance estimation concept but can overcome its limitations. Furthermore, it is shown that the method presented in [3] is a variation of this more general method. The basic idea of the new method is as follows. We can use the sag disturbance to estimate the (fundamental-frequency positive-sequence) impedance of the portion of the system where the sag source does not reside. The sign of the resistive component of this impedance can reveal the location of the disturbance. This conclusion can be proven theoretically for linear systems. Extensive tests on the method using simulation, experiments, and field data have shown that this is a more robust and reliable method.

II. BASIC IDEA OF IMPEDANCE ESTIMATION

The problem of determining sag source or direction can be stated as follows (referring to Fig. 1). A power quality monitor captures sag-containing voltage and current waveforms at point M . It is required to determine if the disturbance comes from the upstream or the downstream. In a typical application, point M is the utility-customer interface point. Ideally, the algorithm of sag source detection should be implemented in a digital revenue meter connected at M . As a result, both the supply utility and customer can obtain a list of disturbances, their severity and directions. Such information will greatly facilitate the resolution of disputes between the two parties if a disturbance results in financial losses to either party.

A. Concept of Impedance Estimation

The system of Fig. 1 can be represented by an equivalent circuit shown in Fig. 2. This is a fundamental-frequency positive-sequence circuit. Z_1 and E_1 are equivalent impedance and

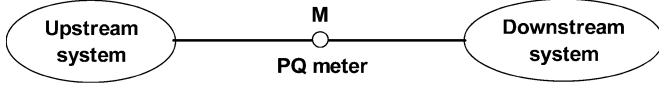


Fig. 1. Problem of locating voltage sag sources.

internal voltage source of the left side (say, supply system). Z_2 and E_2 are the similar parameters for the customer system on the right side. This equivalent circuit model is valid if one assumes that the system consists of linear elements. Point M is the measurement location where voltage and current waveforms are measured.

Assume a disturbance occurs on the customer-side and leads to a voltage sag at M , the measurements at M satisfy this equation before the occurrence of the event

$$V = E_1 - I Z_1. \quad (1)$$

When a disturbance occurs, the voltage and current are changed to $V + \Delta V$ and $I + \Delta I$, where ΔV and ΔI are the voltage and current changes due to the customer-side event. If we assume that the parameters on the supply-side (Z_1 and E_1) remain unchanged during the disturbance period, a similar equation can be written as

$$V + \Delta V = E_1 - (I + \Delta I) Z_1. \quad (2)$$

Since the probability that a disturbance occur on both sides simultaneously is practically zero, the above assumption that the parameters on the nondisturbance side are constant is justifiable.

Subtracting (1) from (2), we can find the impedance of the nondisturbance (supply) side as

$$Z_1 = -\frac{\Delta V}{\Delta I}. \quad (3)$$

Similarly, if a disturbance occurs on the supply-side, the customer-side impedance can be obtained as

$$Z_2 = \frac{\Delta V}{\Delta I}. \quad (4)$$

It can be seen that the quantity $Z_e = \Delta V / \Delta I$ gives different signs depending on the origin of the sag. The basic idea proposed by this paper is, therefore, to estimate Z_e . In fact, Z_e has a physical meaning. It is the equivalent impedance of the nondisturbance side. If the disturbance occurs on the supply-side, Z_e is the customer impedance. If the disturbance occurs on the customer-side, Z_e is the supply impedance multiplied by a negative sign.

Since the resistance should always be positive, we could determine the direction of sag source by checking the sign of the real part of the impedance Z_e . This forms the basis of our proposed method:

- 1) Calculate the equivalent impedance once a voltage sag is detected at monitoring point

$$Z_e = \frac{\Delta V}{\Delta I} = \frac{V_{\text{during}} - V_{\text{pre}}}{I_{\text{during}} - I_{\text{pre}}} \quad (5)$$

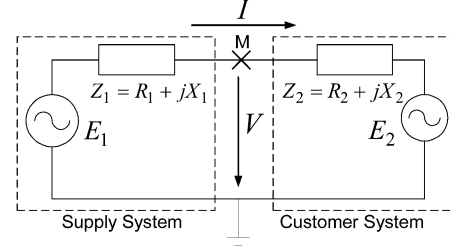


Fig. 2. Equivalent circuit for sag analysis.

where $(V_{\text{pre}}, I_{\text{pre}})$ and $(V_{\text{during}}, I_{\text{during}})$ are pairs of pre-sag and during-sag fundamental-frequency positive-sequence voltage and current, respectively.

- 2) If $\text{Real}(Z_e) > 0$, the source of sag is on supply-side.
- 3) If $\text{Real}(Z_e) < 0$, the source of sag is on customer-side.

Note that the above conclusion is valid with respect to the current direction shown in Fig. 2. In actual implementation, the current direction can be established as the same direction as the active power, e.g., if $\text{Real}(Z_e) > 0$, the disturbance source will be in the upstream direction.

The above method can be graphically illustrated on the impedance plane as shown in Fig. 3. If the calculated impedance Z_e lies in either the first or fourth quadrant ($R_e > 0$), the sag source is on the supply-side. And if the impedance lies in either the second or third quadrant ($R_e < 0$), the sag source is on the customer-side. Since the equivalent impedances of a power network are generally reactive at the fundamental-frequency, impedance results will mostly reside in either the first or third quadrant [6].

III. IMPROVING THE ACCURACY OF IMPEDANCE ESTIMATION

The impedance-based method is theoretically sound, but there is one major difficulty in its practical implementation: the results could be inconsistent if different pre- or during-event cycles are selected. According to our several months' field measurement results, source locations for almost 20% of the disturbances cannot be correctly identified if during-event cycle is selected too close or too far from the pre-event cycle. This is because the estimated impedance would vary during the disturbance period. Therefore, using a single during-event cycle in the analysis could yield unreliable results. In order to improve the estimation of the impedance, we propose to utilize multiple cycles of data by using the least-squares (LS) method. This can be done as follows. Equation (1) can be rewritten as

$$V_X + jV_Y = (E_{1X} + jE_{1Y}) - (I_X + jI_Y)(R_1 + jX_1) \quad (6)$$

where subscripts X and Y represent real and imaginary parts of a variable, respectively. If n cycles of (V, I) data are measured during the evolution of a sag, (6) becomes

$$\begin{bmatrix} V_X(1) \\ \vdots \\ V_X(n) \end{bmatrix} = \begin{bmatrix} I_X(1) & I_Y(1) & 1 \\ \vdots & \vdots & \vdots \\ I_X(n) & I_Y(n) & 1 \end{bmatrix} \begin{bmatrix} -R_1 \\ X_1 \\ E_{1X} \end{bmatrix} \quad (7)$$

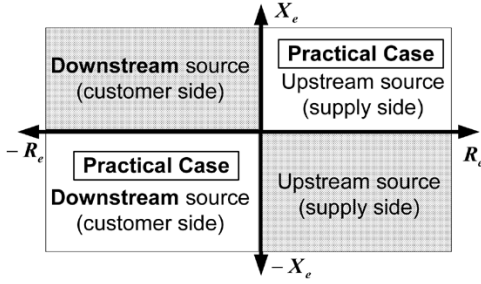


Fig. 3. Impedance plane illustration for result interpretation.

and

$$\begin{bmatrix} V_Y(1) \\ \vdots \\ V_Y(n) \end{bmatrix} = \begin{bmatrix} I_Y(1) & I_X(1) & 1 \\ \vdots & \vdots & \vdots \\ I_Y(n) & I_X(n) & 1 \end{bmatrix} \begin{bmatrix} -R_1 \\ -X_1 \\ E_{1Y} \end{bmatrix}. \quad (8)$$

Most power quality monitors provide **three cycles** or more pre-disturbance and post-disturbance data. Data recorded during the disturbance can be as long as the duration of the disturbance. **As a result, n is typically greater than three.** The equivalent impedance can then be found by using the LS method

$$\begin{bmatrix} -R_1 \\ X_1 \\ E_{1X} \end{bmatrix} = \begin{bmatrix} I_X(1) & I_Y(1) & 1 \\ \vdots & \vdots & \vdots \\ I_X(n) & I_Y(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V_X(1) \\ \vdots \\ V_X(n) \end{bmatrix} \quad (9)$$

and

$$\begin{bmatrix} -R_1 \\ -X_1 \\ E_{1Y} \end{bmatrix} = \begin{bmatrix} I_Y(1) & I_X(1) & 1 \\ \vdots & \vdots & \vdots \\ I_Y(n) & I_X(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V_Y(1) \\ \vdots \\ V_Y(n) \end{bmatrix} \quad (10)$$

where the symbol “+” **indicates pseudo-inverse [7]**. From (9) and (10), it can be seen that the impedance calculated from the right side will always have a negative real part. This implies that for a customer-side disturbance, if we try to find impedance by calculating the right side of (9) or (10), the result will drop into the second or third quadrant of the impedance plane ($R_e < 0$) as shown in Fig. 3.

Similarly, if the disturbance is originated from the supply-side one obtains

$$\begin{bmatrix} R_2 \\ -X_2 \\ E_{2X} \end{bmatrix} = \begin{bmatrix} I_X(1) & I_Y(1) & 1 \\ \vdots & \vdots & \vdots \\ I_X(n) & I_Y(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V_X(1) \\ \vdots \\ V_X(n) \end{bmatrix} \quad (11)$$

and

$$\begin{bmatrix} R_2 \\ X_2 \\ E_{2Y} \end{bmatrix} = \begin{bmatrix} I_Y(1) & I_X(1) & 1 \\ \vdots & \vdots & \vdots \\ I_Y(n) & I_X(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V_Y(1) \\ \vdots \\ V_Y(n) \end{bmatrix}. \quad (12)$$

Therefore, it can be concluded that for a supply-side disturbance, the impedance found from (11) or (12) will lie in the first or fourth quadrant of the impedance plane ($R_e > 0$) as shown in Fig. 3. The calculated impedance will always have a positive real part. **Note the terms on the right of (11) and (12) are exactly the same as that of (9) and (10), respectively.**

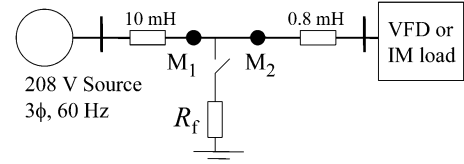


Fig. 4. Test circuit for constant power loads.

Accordingly, the source direction of sag can be detected by checking the sign of real part of the impedance calculated by the proposed LS-based method. Positive real part of the fundamental-frequency impedance indicates that the sag is originated from the supply-side, negative one implies it is on the customer-side.

The proposed method can be implemented in the following steps.

- 1) Perform Fourier analysis to extract the fundamental-frequency positive-sequence voltage and current **cycle-by-cycle** during the voltage sag event.
- 2) Separate real and imaginary parts of voltage and current and calculate the values of R_e from

$$\begin{bmatrix} R_e \\ X_e \\ E_X \end{bmatrix} = \begin{bmatrix} I_X(1) & I_Y(1) & 1 \\ \vdots & \vdots & \vdots \\ I_X(n) & I_Y(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V_X(1) \\ \vdots \\ V_X(n) \end{bmatrix} \quad (13)$$

and

$$\begin{bmatrix} R_e \\ X_e \\ E_Y \end{bmatrix} = \begin{bmatrix} I_Y(1) & I_X(1) & 1 \\ \vdots & \vdots & \vdots \\ I_Y(n) & I_X(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V_Y(1) \\ \vdots \\ V_Y(n) \end{bmatrix}. \quad (14)$$

- 4) Check the consistency of the resistance signs determined from (13) and (14). **If these two signs are not consistent (opposite), no conclusion can be drawn.**
- 5) If the signs are consistent, then positive signs indicate that the source is on the supply-side and negative signs indicate the source is on the customer-side.

IV. NONLINEAR OR CONSTANT POWER LOADS

Another issue that could render the proposed method less reliable is the assumption of linear system. In reality, there often exists nonlinear loads such as variable frequency drives (VFDs) and induction motors (IMs) in the customer facilities. **Both types of loads could behave as constant power loads** [8], [9]. Their response during the voltage sag can be quite different from the linear loads. In order to clarify this issue, experimental studies were conducted. Results are presented in this section.

A. Experimental Setup

A laboratory experiment consisting of a VFD or an IM load was set up for this purpose (Fig. 4). The VFD or IM has a rating of 2 HP. Various voltage sags ranging from 5% to 20% drop were introduced by switching R_f in and out of the circuit. The sag started at the 0th cycle and lasted for about 10 cycles. Voltage and current signals were recorded at two points— M_1 and M_2 —giving downstream (DS) and upstream (US) sag-source data, respectively. Fig. 5 shows voltage and current waveforms for the case of 20% sag.

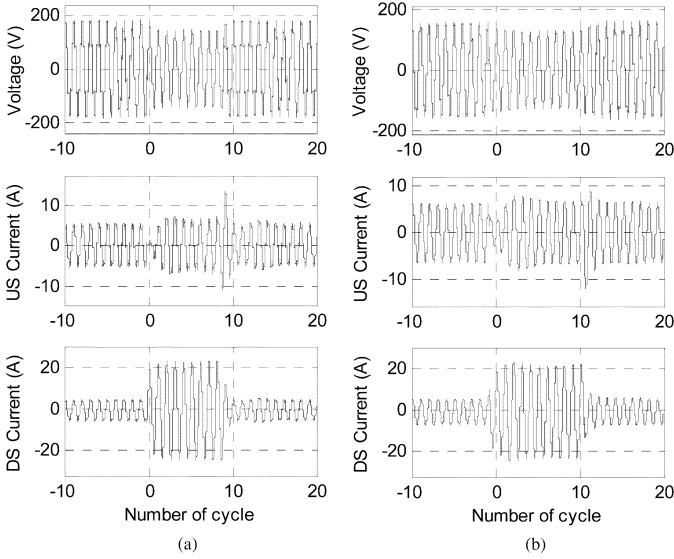


Fig. 5. Voltage and current waveforms. (a) VFD load. (b) IM load.

Since the supply system is dominated by linear components, the proposed method has no difficulties to identify downstream sag sources. The difficulties arise when a sag is originated from the upstream. **The proposed method essentially attempts to estimate an equivalent impedance for the downstream which is a constant power load. The upstream disturbance is therefore the focus of this investigation.** Fig. 6 shows the fundamental-frequency positive-sequence voltage, current and powers associated with an upstream sag. **It can be seen that the active power and current decrease initially and then increase in order to compensate the drop in the voltage for maintaining the constant power.** Such a behavior has been well investigated in the voltage stability area [10], [11]. According to [11], the response can be represented using a dynamic load model with transient and steady-state load characteristic curves.

The transient characteristic curves have the form of $P_t(V) = P_0 V^\alpha$ and $Q_t(V) = Q_0 V^\beta$ and are shown as dotted lines in the last row of Fig. 6. The range of α is estimated to be 4.2 to 6.5 for a VFD and 11.0 to 16.1 for an IM. The range of β varies from 10.6 to 17.9 for a VFD and 3.5 to 4.5 for an IM. The transient response of the loads is, therefore, not the constant power ($\alpha = \beta = 0$) type. In fact, the load is much more sensitive to voltage change than the linear load of $\alpha = \beta = 2$. Thus, if Z_e for the transient period is estimated, one may detect the sag source reliably. The theoretical justification for this approach is provided as follows. The load can be modeled as

$$P + jQ = P_0 V^\alpha + jQ_0 V^\beta. \quad (15)$$

The corresponding load current is

$$I = \frac{P - jQ}{V} = P_0 V^{\alpha-1} - jQ_0 V^{\beta-1}. \quad (16)$$

In the previous equation, the phase angle of voltage V is used as the reference angle and is set to zero. If there is a voltage disturbance ΔV , the current change ΔI is

$$\Delta I = P_0 V^{\alpha-2}(\alpha - 1)\Delta V - jQ_0 V^{\beta-2}(\beta - 1)\Delta V. \quad (17)$$

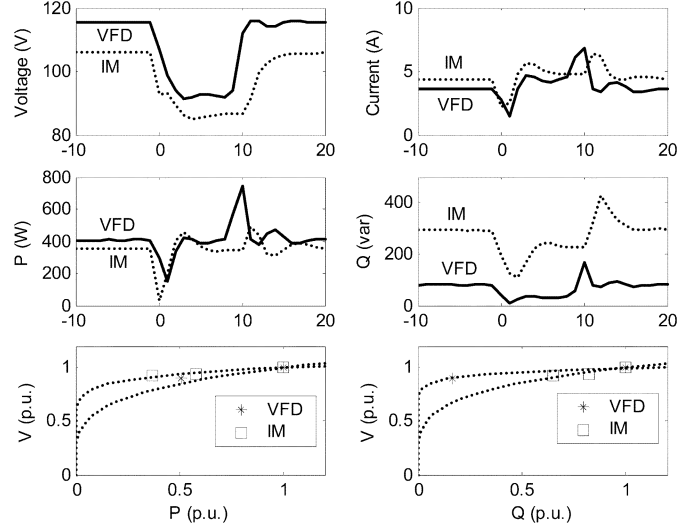


Fig. 6. Fundamental positive-sequence voltage, current, power, and characteristics of voltages against power.

The equivalent incremental impedance of the load can thus be estimated as

$$\begin{aligned} \Delta Z &= \frac{\Delta V}{\Delta I} = \frac{1}{P_0 V^{\alpha-2}(\alpha - 1) - jQ_0 V^{\beta-2}(\beta - 1)} \\ &= k P_0 V^{\alpha-2}(\alpha - 1) + jk Q_0 V^{\beta-2}(\beta - 1) \\ &= R_{eq} + jX_{eq} \end{aligned} \quad (18)$$

where $k = [P_0 V^{\alpha-2}(\alpha - 1)]^2 + [Q_0 V^{\beta-2}(\beta - 1)]^2$. R_{eq} is the resistance to be determined by the sag source detection algorithm. The resistance has the form of

$$R_{eq} = k P_0 V^{\alpha-2}(\alpha - 1). \quad (19)$$

It can be seen that if $\alpha > 1$, R_{eq} will be positive. So the proposed method will work. Since α is much larger than 1, R_{eq} will be larger and it will make the sign determination easier.

B. Modified Sag-Source Detection Method

In order to determine how many cycles of data are sufficient to capture the transient characteristics of the load, a sensitivity study has been conducted. Fig. 7 shows the results that are R_e values calculated using different window lengths. The figures also plot the active power change defined as $\Delta P = P - P_{pre}$ where P is (cycle-by-cycle) active power. This index can be used to monitor the power restoration behavior of a constant power load.

It can be seen that, if the window is less than three cycles, the two estimated R_e values are close to each other and their signs indicate sag locations correctly. The active power index starts to change sign after three cycles, which means that the load power has restored after three cycles and the load no longer behaves as a linear load. Based on these observations and the inherent transient characteristics of the loads, the sag-source detection technique is further refined. The basic idea is to select the during-sag cycle properly for Z_e estimation. The criterion is to monitor the active power change with respect to its pre-disturbance value. **If ΔP has only one sign, either positive or negative, the value of n for Z_e estimation can be selected as the sag duration. If ΔP changes sign, the value of n should be selected as $K - 1$, where**

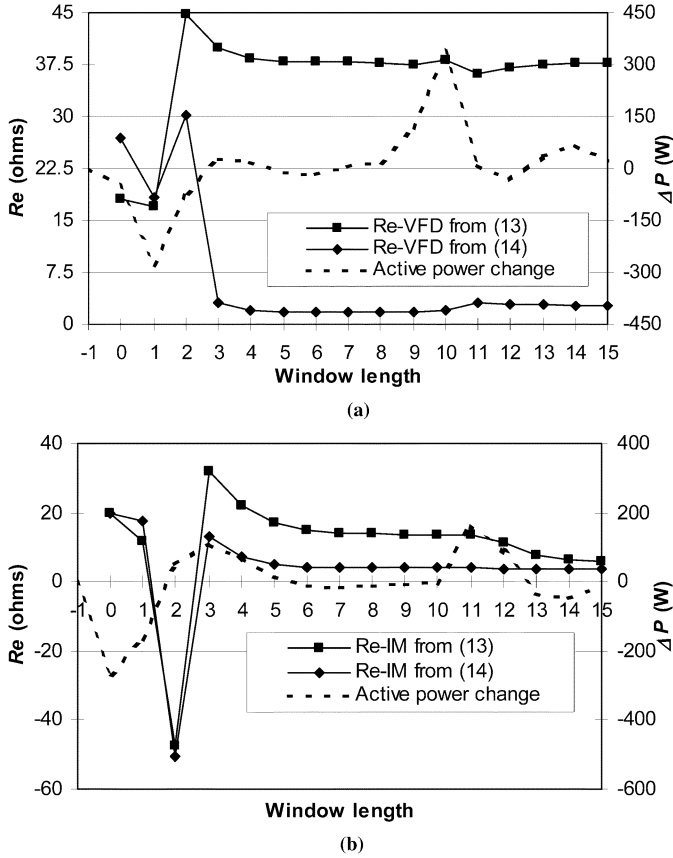


Fig. 7. Resistance plots using different window length. (a) VFD load. (b) IM load.

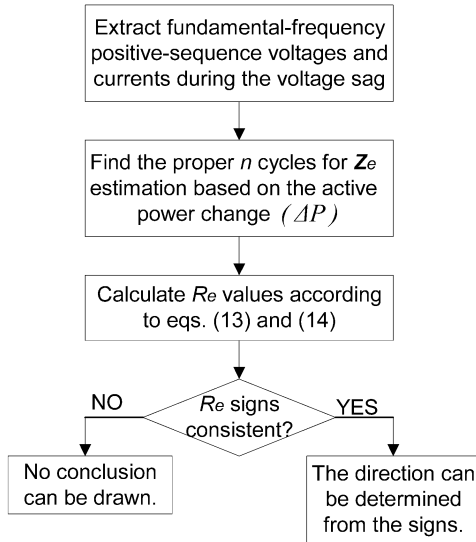


Fig. 8. Flowchart of the proposed method.

K is the cycle when ΔP changes sign. The final algorithm of the proposed sag source detector is thus shown in Fig. 8.

The above method assumes that a composite customer load has transient power recovery characteristic. Extensive voltage stability oriented dynamic load research and measurement such as those shown in [12] and [13] supports the validity of this assumption. It is theoretically possible that some power electronic load could respond faster than three cycles. Our extensive literature search on this subject has not found the existence of such

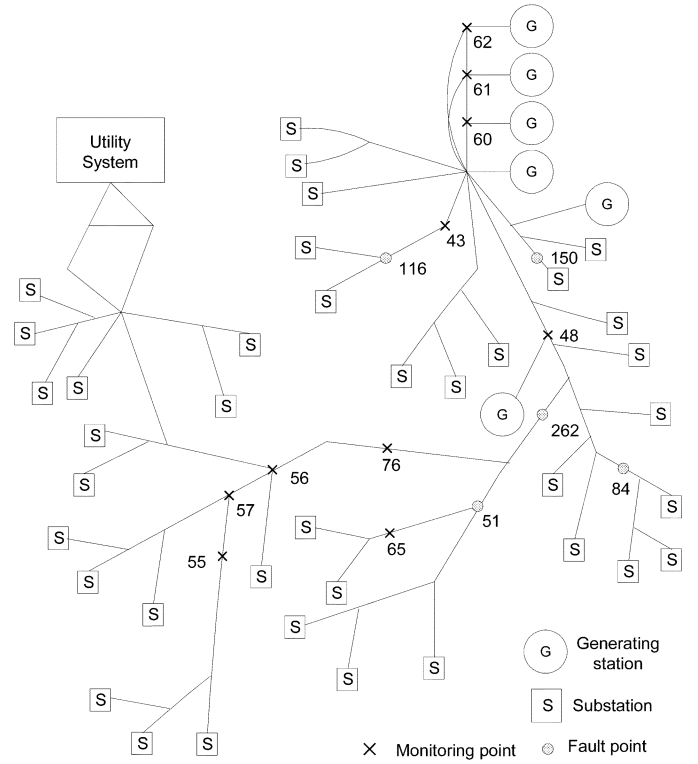


Fig. 9. Simplified system diagram for the simulation.

TABLE I
BUS SELECTION DETAILS AND RESISTANCE VALUES

Fault Bus	Monitoring buses		The average R_f values (p.u.)	
	DSF	USF	DSF	USF
262	61	57	-0.12	+1.55
150	62	76	-0.15	+1.31
84	43	55	-0.08	+4.25
51	48	56	-0.07	+4.04
116	60	65	-0.04	+3.73

very fast response characteristic. It is important to note as well that sag source detection methods are not intended for application to individual power electronic loads. They are intended for the customer-utility interface points. The downstream loads are composite loads representing a whole customer facility. It is an accepted fact that such loads exhibit recovery characteristics.

V. VERIFICATION STUDIES

The proposed method has been verified using simulation, field measurement and lab experiment results.

A. Simulation Results

Three-phase-to-ground faults causing balanced sags are simulated at five different locations in a 100-bus power system. Each fault/sag event is recorded at two buses, one at the downstream of the fault (DSF) and the other at the upstream (USF). The studying system diagram is shown in Fig. 9. The fundamental-frequency resistance (R_1) values averaged from three phases are tabulated in Table I. It can be seen that signs of resistance can indicate the sag sources (or fault bus directions) correctly.

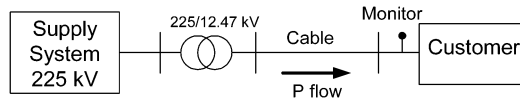


Fig. 10. One-line diagram of the field measurement system.

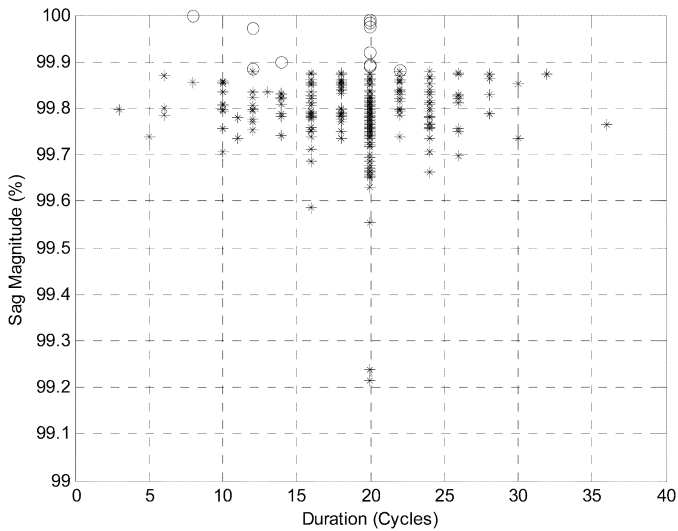


Fig. 11. Magnitude of voltage sags as a function of their duration.

B. Field Measurement Results

The proposed method has also been tested using field data. A power quality monitor is placed at a 12.47-kV industrial plant which is fed from a 225-kV substation as shown in Fig. 10. **Both the supply and customer-sides have power generations.** The plant consists of two groups of variable frequency drives (VFDs), the fifth, and seventh harmonic filters installed at 12.47-kV bus, and five local generators supplying six additional VFDs and commercial loads such as office-building loads and heating. Disturbance waveforms were monitored continuously for several months and lots of voltage sags were detected. Historical record and operation log show that the disturbances were mainly due to heavy load-switchings causing sags on the customer-side. Fig. 11 shows a scatter plot of voltage sag magnitudes versus their duration. It can be seen that most sags are not severe and the duration is around 15–25 cycles. The circles in the figure are the disturbances whose direction cannot be detected by the proposed method due to inconsistent signs of the estimated R_e values.

Captured events are analyzed by calculating the equivalent impedance from (5) at first. For all events, the first cycle before voltage drop and the last cycle of an event are selected for impedance calculations. The results are plotted in an impedance plane as shown in Fig. 12. Note that each impedance value is associated with the current change (ΔI) range in A. **Reliable results can be achieved when ΔI values are high.** It can be seen that most impedances lie in the third quadrant, which indicates that the sources of these sags are from the customer-side. The negative values of these impedances are supposed to be the driving-point impedances of the supply system (Z_1 in Fig. 2). However, there are still some impedances located in the fourth quadrant, which could be wrongly interpreted as supply-side disturbances. They cannot be trusted, however, if we examine the values of those impedances further. The points in the fourth

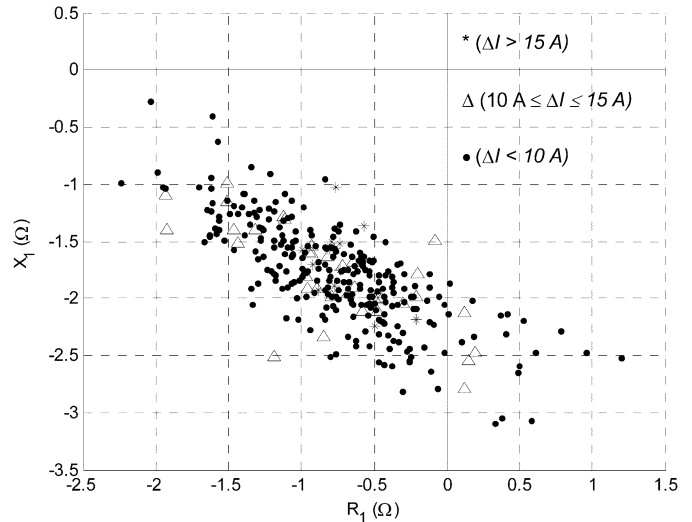


Fig. 12. Impedances estimated using two cycles of data.

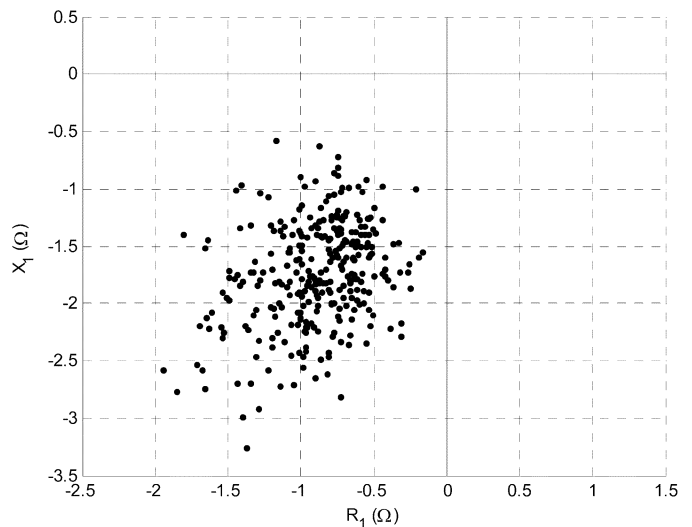


Fig. 13. Impedances estimated using the proposed least-squares method.

quadrant are supposed to be the equivalent fundamental-frequency impedance of the customer-side Z_2 , but their values are close to the supply system driving-point impedances as plotted in the third quadrant, in addition, the impedances are capacitive. This is impossible in a practical distribution system. Therefore, it is very likely that all impedances should lie in the third quadrant. **The errors are caused by the improper selection of pre- and during-event cycles.**

The results can be significantly improved by using more data cycles in the proposed **least-squares method**, as shown in Fig. 13. The average of two consistent impedance values [one from (13) and the other one from (14)] is used to represent a single impedance value. **Events with inconsistent resistance signs are considered as nondetectable by the proposed method. Their R_e results are not plotted in Fig. 13.** But the magnitude and duration of the disturbances are shown in Fig. 11 as circles. It can be seen that such disturbances have very **small** magnitudes. Failure to detect their direction is caused by lower energy levels of the disturbance and should not be a major concern, since they are not significant events. It can be seen from Fig. 13 that all the impedances now are only located in the third quadrant, which indicates that they are generated by customer-side

TABLE II
EXPERIMENTAL RESULTS

Event	Source location	Sag type	Sag level (%)	Duration (cycles)	$Re (\Omega)$
VARIABLE FREQUENCY DRIVE (VFD)					
1	Downstream	3-P	4.1	12	-0.1
2	Downstream	3-P	21.7	10	-0.1
3	Upstream	3-P	4.1	12	+12.3
4	Upstream	3-P	21.7	10	+18.3
5	Upstream	3-P	3.4	120	+22.0
6	Upstream	3-P	8.6	120	+22.2
7	Upstream	3-P	13.8	120	+20.3
8	Upstream	3-P	20.2	120	+24.9
9	Upstream	1-P	2.9	120	+14.5
10	Upstream	1-P	8.3	120	+16.8
11	Upstream	1-P	17.2	120	+24.5
12	Upstream	1-P	18.5	120	+20.7
INDUCTION MOTOR (IM)					
1	Downstream	3-P	19.8	13	-0.1
2	Upstream	3-P	19.8	13	+14.2
3	Upstream	3-P	2.9	120	+18.8
4	Upstream	3-P	8.1	120	+21.9
5	Upstream	3-P	13.9	120	+14.8
6	Upstream	3-P	20.1	120	+18.2
7	Upstream	1-P	2.1	120	+23.7
8	Upstream	1-P	6.0	120	+19.1
9	Upstream	1-P	10.1	120	+11.2
10	Upstream	1-P	14.4	120	+18.0
LINEAR LOAD					
1	Downstream	3-P	4.2	12	-0.2
2	Upstream	3-P	4.2	12	+31.1
3	Upstream	3-P	4.4	7	+31.4
4	Upstream	3-P	3.9	18	+31.0

disturbances. Their values are also more concentrated on the real supply-side driving-point impedance, $0.78+j1.70 \Omega$ which is derived from the detailed system model. The field test results show great robustness and improvement by using the proposed method, this is mainly due to the employment of redundant information.

C. Experimental Results

The proposed method is also evaluated using experimental studies. The experimental set up shown in Fig. 4 was used to produce a number of sags include upstream sags and unbalanced sags. Table II summarizes results of sag source detection for VFD, IM and linear loads, which further confirm that the proposed method can work reliably for both linear and nonlinear loads.

VI. DISCUSSION

This section discusses practical issues associated with the proposed method such as swells and unbalanced sags, the method simplification, applications for short-duration sags and meshed systems and errors of instruments.

A. Swells and Unbalanced Sags

Unbalanced sags refer to those sags that exhibit different sag levels for different phases. One or two phases may even experience voltage swell. The proposed method applies to swells and unbalanced sags equally well. This is because the estimation of the equivalent positive-sequence impedance is not dependent

on the disturbance type. As long as there is a disturbance, the impedance can be estimated.

B. Simplification of the Method

Since the proposed method examines the sign of resistance only, there is no need to calculate the reactance. The impedance can be decoupled by introducing rotating transformation, that is, selecting the fundamental-frequency current vector as reference axis. If we multiply $e^{-j\alpha}$ on both sides of (6), where $\alpha = \tan^{-1}(I_Y/I_X)$ is the phase angle of the measured current, then (7) and (8) can be simplified as

$$\begin{bmatrix} V \cos \theta(1) \\ \vdots \\ V \cos \theta(n) \end{bmatrix} = \begin{bmatrix} I(1) & 1 \\ \vdots & \vdots \\ I(n) & 1 \end{bmatrix} \begin{bmatrix} -R \\ E_1 \cos \theta_1 \end{bmatrix} \quad (20)$$

and

$$\begin{bmatrix} V \sin \theta(1) \\ \vdots \\ V \sin \theta(n) \end{bmatrix} = \begin{bmatrix} I(1) & 1 \\ \vdots & \vdots \\ I(n) & 1 \end{bmatrix} \begin{bmatrix} -X \\ E_1 \sin \theta_1 \end{bmatrix} \quad (21)$$

where θ and θ_1 are the phase angles by which V and E_1 lead I , respectively. In this transformation, it is assumed that θ_1 will not change significantly during the event. The equivalent resistance R can be obtained as follows by using the LS technique:

$$\begin{bmatrix} -R \\ E_1 \cos \theta_1 \end{bmatrix} = \begin{bmatrix} I(1) & 1 \\ \vdots & \vdots \\ I(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V \cos \theta(1) \\ \vdots \\ V \cos \theta(n) \end{bmatrix} \quad (22)$$

Similarly, the parameters on customer-side can be found from a supply-side event as

$$\begin{bmatrix} R \\ E_2 \cos \theta_2 \end{bmatrix} = \begin{bmatrix} I(1) & 1 \\ \vdots & \vdots \\ I(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V \cos \theta(1) \\ \vdots \\ V \cos \theta(n) \end{bmatrix} \quad (23)$$

where θ_2 is the phase by which E_2 leads I , θ_2 is also assumed to be constant during the disturbance.

It turns out that the above equation is very similar to that of [3]. Thus a theoretical foundation is established for the method in [3]. Note that, however, the method in [3] assumes θ_1 and θ_2 do not change significantly during a disturbance. This might not be true under some practical conditions. The method proposed in this paper, however, does not rely on this assumption and is more reliable.

C. Applications for Short-Duration Sags and Meshed Systems

From (22), it can be seen that theoretically at least two sampling points during the event, among which one point is pre-disturbance measurement, are required for resistance estimation. Since the majority of voltage sags last several cycles or longer, this requirement can be easily satisfied. If there are indeed only two data points, the results can be unreliable. This is a major drawback of the proposed method. However, the high frequency components contained in the waveforms of such sags could be

used to improve the reliability of impedance estimation. We are currently investigating this subject.

The proposed method is developed for radial systems where upstream and downstream subsystems can be identified clearly. The interface point between a utility and a customer belongs to this type and the method is reliable to determine the sag responsibility. This is the most useful and the intended application of the proposed method. **The method may not be reliable for meshed systems.** More research work is needed to confirm this suspicion.

D. Errors of Instruments

Measurement errors can be divided into three main groups: instrument errors, transducer errors and errors due to the measured signal (low level signals). In practice, the main errors are from the recording instrument, voltage and current transformers (PTs and CTs). The proposed method works on the fundamental-frequency components. Errors from these sources are very small at the fundamental-frequency. Modern power quality meters generally possess sufficient sampling rate for the fundamental-frequency analysis. In summary, the proposed method does not impose additional accuracy requirements. The instrumentation circuits of modern digital revenue meters are sufficient for reliable implementation of the proposed method.

VII. CONCLUSION

The sign of the fundamental-frequency positive-sequence resistance estimated at a metering point is proposed as an indicator to determine the direction of voltage sag source. The resistance is derived using the least-squares method. Main conclusions of this paper are summarized as follows.

- 1) The principle of the proposed method is to estimate the equivalent impedance of the nondisturbance side by utilizing the voltage and current changes caused by the disturbance.
- 2) It requires only measuring the voltage and current waveforms. The fundamental-frequency positive-sequence components of the waveforms are extracted for impedance estimation. As a result, the method can be easily implemented in modern digital revenue meters.
- 3) The proposed method has a solid theoretical foundation for linear systems and does not rely on empirical assumptions. For nonlinear loads, the method has been improved by taking into account the transient characteristics of the load.

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